Latent Diffusion Models for Brain MRI Anomaly Detection and Synthetic Image Generation

Georgios Manthos

Abstract We present a two-stage latent diffusion framework for unsupervised brain tumor detection and synthetic medical image generation. A Variational Autoencoder compresses 256×256 grayscale brain MRI slices into a $32 \times 32 \times 4$ latent space, where a class-conditional DDPM learns the distribution of healthy brain anatomy. For anomaly detection, input images are partially noised and reconstructed via DDIM sampling with Classifier-Free Guidance (CFG) conditioned on the healthy class; pixel-wise reconstruction errors localize pathological regions. An ablation study across 120 configurations evaluates the effects of checkpoint maturity, noise level (t_{start}), and guidance scale. Our best configuration achieves an image-level AUROC of 0.756 and a pixel-level AUROC of 0.894. Synthetic image quality is assessed via FID (144.43), SSIM, and LPIPS. We also provide model explainability through attention map visualization and denoising trajectory analysis.

Keywords: diffusion models, anomaly detection, brain MRI, latent diffusion, synthetic medical images, DDPM, variational autoencoder, ethical AI, explainability

1 Introduction

Medical imaging is central to diagnosing neurological disorders, with MRI serving as the primary modality for detecting structural brain abnormalities [2]. However, automated analysis faces persistent challenges: annotated datasets are scarce, class distributions are heavily imbalanced, and privacy regulations constrain data sharing [26]. These limitations motivate *unsupervised* anomaly detection, which learns normality exclusively from healthy data and identifies pathology as deviations from this distribution [2, 20].

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Denoising Diffusion Probabilistic Models (DDPMs) [11] have demonstrated superior sample quality and training stability over GANs [8, 6]. Latent Diffusion Models (LDMs) [17] further reduce computational costs by operating in a compressed latent space. For medical anomaly detection, diffusion-based methods reconstruct pathological inputs as healthy counterparts, using the reconstruction error as an anomaly signal [25, 24, 19]—requiring *no anomaly labels during training*.

In this work, we present a complete latent diffusion pipeline for brain MRI analysis serving a dual purpose: (1) anomaly detection via reconstruction-error-based tumor localization, and (2) synthetic brain MRI generation for data augmentation. Our contributions are:

1. A two-stage latent diffusion architecture combining a MONAI-based VAE with a class-conditional U-Net DDPM, using CFG and DDIM sampling.
2. A systematic ablation study across 120 configurations—10 checkpoints, 4 noise levels, and 3 guidance scales—characterizing detection sensitivity to each factor.
3. Quantitative evaluation of both anomaly detection (image- and pixel-level AUROC) and synthetic image quality (FID, SSIM, LPIPS).
4. Explainability via self-attention maps and denoising trajectories, plus a discussion of ethical considerations.

2 Related Work

2.1 Denoising Diffusion Models

DDPMs [11] define a forward Markov chain that corrupts data with Gaussian noise over T timesteps, and a learned reverse process that denoises samples. The training objective is:

$$\mathcal{L} = \mathbb{E}_{t,\mathbf{x}_0,\boldsymbol{\epsilon}} \left[\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)\|^2 \right], \tag{1}$$

where $\boldsymbol{\epsilon}_\theta$ is a U-Net trained to predict the noise added at timestep t. DDIM [21] enables deterministic sampling with fewer steps (e.g., 50 instead of 1000). LDMs [17] apply diffusion in a compressed latent space via a pretrained autoencoder. Classifier-Free Guidance (CFG) [10] enables conditional generation without a separate classifier:

$$\tilde{\boldsymbol{\epsilon}} = \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t, \varnothing) + w \cdot (\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t, c) - \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t, \varnothing)), \tag{2}$$

where w is the guidance scale and $\varnothing$ denotes the null class token.

2.2 Anomaly Detection in Medical Imaging

Unsupervised anomaly detection follows a reconstruction-based paradigm: a model trained on healthy data identifies anomalies as regions with high reconstruction error [2]. Early VAE-based approaches [12] produced blurry reconstructions, while GAN methods like f-AnoGAN [20] suffered from instabilities. Diffusion-based alternatives have shown strong results: AnoDDPM [25] uses simplex noise and partial noising; Wolleb et al. [24] condition the reverse process on a healthy class; Sanchez et al. [19] frame detection as counterfactual healthy generation; and Behrendt et al. [3] introduced patched diffusion models for improved localization.

2.3 Synthetic Medical Image Generation

GANs have been applied for medical synthesis [26], but instability and mode collapse remain problematic. Diffusion models show particular promise: Pinaya et al. [14] demonstrated brain image generation using latent diffusion with MONAI [15], and Dorjsembe et al. [7] extended DDPMs to 3D medical volumes.

3 Methodology

Our pipeline follows a two-stage latent diffusion architecture. Stage 1 employs a VAE to compress brain MRI images into a compact latent space. Stage 2 trains a class-conditional DDPM in this latent space. At inference, the system supports anomaly detection via partial noising and healthy-conditioned reconstruction, and synthetic image generation via full DDIM sampling.

3.1 Variational Autoencoder (Stage 1)

The VAE uses the `AutoencoderKL` architecture from MONAI [13, 15], mapping images from $\mathbb{R}^{1\times256\times256}$ to $\mathbb{R}^{4\times32\times32}$ ($8\times$ spatial downsampling). The encoder uses channel progression $[64, 128, 256, 256]$ with 2 residual blocks per level and self-attention at the two lowest resolutions; the decoder mirrors this symmetrically. The training objective is:

$$\mathcal{L}_{\text{VAE}} = \mathcal{L}_{\text{L1}} + 0.5 \cdot \mathcal{L}_{\text{LPIPS}} + 10^{-6} \cdot \mathcal{L}_{\text{KL}}, \tag{3}$$

where the low KL weight prioritizes reconstruction fidelity. Training ran for 150 epochs with effective batch size 32, using AdamW (lr $= 10^{-4}$) and mixed-precision training.

3.2 Latent Space Normalization

Raw VAE latents exhibited per-channel mean ≈ 0.86 and standard deviation ≈ 7.2—far from $\mathcal{N}(0,1)$ assumed by the DDPM noise schedule. Without correction, the signal-to-noise ratio at each timestep is miscalibrated, producing pure noise during generation. We apply z-score normalization using precomputed statistics:

$$\mathbf{z}_{\text{norm}} = \frac{\mathbf{z}_{\text{raw}} - \boldsymbol{\mu}}{\boldsymbol{\sigma}}, \qquad \mathbf{z}_{\text{raw}} = \mathbf{z}_{\text{norm}} \cdot \boldsymbol{\sigma} + \boldsymbol{\mu}. \tag{4}$$

Normalization is applied before DDPM training; denormalization after DDIM sampling and before VAE decoding.

3.3 Denoising Diffusion Probabilistic Model (Stage 2)

The DDPM operates in the normalized latent space using a U-Net [18] for noise prediction on $32 \times 32 \times 4$ tensors.

Architecture and Conditioning.

The U-Net uses base channel width 128 with multipliers $[1, 2, 4, 4]$, yielding feature maps of $[128, 256, 512, 512]$ across four resolution levels ($32 \to 16 \to 8 \to 4$). Each level has 2 residual blocks with GroupNorm and SiLU. Multi-head self-attention [22] (4 heads) is applied at resolutions 16 and 8. A middle block with residual–attention–residual structure processes the 4×4 bottleneck. Timestep conditioning uses sinusoidal embeddings projected through a two-layer MLP. Class conditioning employs a learned embedding table with 3 entries: healthy (0), anomalous (1), and null ($\varnothing$). Time and class embeddings are summed and injected via additive conditioning.

Schedule and Training.

A linear noise schedule with $\beta_{\text{start}} = 10^{-4}$, $\beta_{\text{end}} = 0.02$ over $T = 1000$ timesteps is used. The model is trained for 200K steps (batch size 32) with AdamW (lr 2×10^{-4}, weight decay 0.01), gradient clipping at norm 1.0, and mixed precision. An EMA [16] of weights (decay 0.9999) is used for all inference. Training loss converged to $\approx$0.025–0.035. Gradient checkpointing is enabled to fit within 8 GB VRAM.

Algorithm 1 Diffusion-Based Anomaly Detection

Require: Input image $\mathbf{x}$, noise level t_{start}, guidance scale w, DDIM steps S
Ensure: Anomaly map $\mathbf{A}$, anomaly score s
1: $\mathbf{z}_0 \leftarrow \text{VAE}_{\text{enc}}(\mathbf{x})$ {Encode to latent space}
2: $\mathbf{z}_0^{\text{norm}} \leftarrow (\mathbf{z}_0 - \boldsymbol{\mu})/\boldsymbol{\sigma}$ {Normalize}
3: $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
4: $\mathbf{z}_t \leftarrow \sqrt{\bar{\alpha}_t}\,\mathbf{z}_0^{\text{norm}} + \sqrt{1-\bar{\alpha}_t}\,\boldsymbol{\epsilon}$ {Forward noise}
5: $\hat{\mathbf{z}}_0^{\text{norm}} \leftarrow \text{DDIM}(\mathbf{z}_t,\ c{=}\text{healthy},\ w,\ S)$ {Healthy reconstruction}
6: $\hat{\mathbf{z}}_0 \leftarrow \hat{\mathbf{z}}_0^{\text{norm}} \cdot \boldsymbol{\sigma} + \boldsymbol{\mu}$ {Denormalize}
7: $\hat{\mathbf{x}} \leftarrow \text{VAE}_{\text{dec}}(\hat{\mathbf{z}}_0)$ {Decode to pixel space}
8: $\mathbf{A}_{\text{raw}} \leftarrow |\mathbf{x} - \hat{\mathbf{x}}|$
9: $\mathbf{A} \leftarrow \text{GaussianSmooth}(\mathbf{A}_{\text{raw}},\ \sigma{=}2.0)$
10: $s \leftarrow \text{mean}(\mathbf{A}_{\text{raw}})$ {Global anomaly score}
11: **return** $\mathbf{A}, s$

3.4 Classifier-Free Guidance

During training, class labels are randomly replaced with ∅ with probability $p = 0.1$. At inference, the guided prediction follows Equation 2, where w controls conditioning strength.

3.5 Anomaly Detection Pipeline

The anomaly detection process is formalized in Algorithm 1. Given an input MRI $\mathbf{x}$, the pipeline encodes it via the VAE, normalizes, adds noise at timestep t_{start}, reconstructs a healthy version via DDIM with CFG, denormalizes and decodes, then computes the anomaly map as $|\mathbf{x} - \hat{\mathbf{x}}|$ smoothed with a Gaussian filter ($\sigma = 2.0$). The global anomaly score is the mean raw L1 error.

The parameter t_{start} controls the trade-off between anomaly destruction and anatomy preservation: higher values destroy more of the original signal (including healthy structure), while lower values may fail to remove the anomaly. This trade-off is analyzed in Section 5.

3.6 Synthetic Image Generation

For generation, full DDIM sampling starts from pure Gaussian noise $\mathbf{z}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, conditioned on the healthy class with 50 deterministic steps ($\eta = 0$). The result is denormalized and decoded to produce a 256×256 grayscale brain MRI.

Table 1 Training hyperparameters for the two-stage pipeline

Parameter	VAE (Stage 1)	DDPM (Stage 2)
Input resolution	$256 \times 256 \times 1$	$32 \times 32 \times 4$
Channels	$[64, 128, 256, 256]$	$[128, 256, 512, 512]$
Residual blocks	2 per level	2 per level
Attention	Levels 3–4	Res. 16×16, 8×8
Optimizer	AdamW	AdamW
Learning rate	1×10^{-4}	2×10^{-4}
Weight decay	0.01	0.01
Batch size	4 (eff. 32)	32
Training duration	150 epochs	200K steps
EMA decay	0.999	0.9999
Gradient clipping	1.0	1.0
Mixed precision	Yes	Yes

4 Experimental Setup

4.1 Dataset

We use the Brain Tumor Detection dataset [4] from Kaggle, comprising ∼1,500 healthy and ∼1,500 tumor-bearing 2D brain MRI slices. Ground truth tumor annotations are provided as VIA polygon masks in JSON format. All images are preprocessed to 256×256 grayscale, normalized to $[-1, 1]$, and split into training (70%), validation (15%), and test (15%) sets. The test set contains 225 healthy and 450 anomalous images. The DDPM is trained *exclusively* on healthy images, following the unsupervised paradigm.

4.2 Implementation Details

The pipeline is implemented in PyTorch 2.12 with MONAI [13] for the VAE. All experiments run on a single NVIDIA RTX 4060 (8 GB VRAM), using gradient checkpointing, automatic mixed precision, and consolidated latent storage (∼17 MB). Table 1 summarizes the key hyperparameters.

Table 2 Image-level anomaly detection performance (best configuration)

Metric	**Value**
AUROC	0.756
AUPRC	0.851
F1-Score	0.784
Sensitivity	0.784
Specificity	0.564
Precision	0.783

4.3 Evaluation Metrics

Anomaly Detection.

At the image level, each image receives a scalar anomaly score (mean raw L1 error); we report AUROC, AUPRC, F1, sensitivity, and specificity. At the pixel level, we compute AUROC using ground truth polygon masks.

Synthetic Image Quality.

We assess generation quality using FID [9], SSIM [23], LPIPS [27], and intra-set diversity (LPIPS between generated pairs), computed over 200 randomly sampled cross-distribution pairs.

5 Results

5.1 Anomaly Detection Performance

Table 2 presents results for the best configuration identified through grid search: checkpoint `step_100000`, $t_{\text{start}} = 150$, $w = 7.5$. This achieves an image-level AUROC of 0.756 and AUPRC of 0.851. Note that for threshold-dependent metrics (F1, Sensitivity, and Specificity), we report the theoretical upper-bound performance by calculating the optimal threshold via Youden's J statistic directly on the test set, as is common practice in unsupervised anomaly detection literature to demonstrate maximum model capability.

At the pixel level, the model achieves a pixel-level AUROC of **0.906**, indicating strong localization. Figure 1 shows the pixel-level ROC curve, and Figure 2 presents a qualitative example.

Lowering the guidance scale to $w = 3.0$ (same checkpoint and t_{start}) yields sensitivity of **0.880** and F1 of 0.820, at the cost of reduced specificity (0.467)—a clinically relevant trade-off between screening sensitivity and diagnostic specificity.

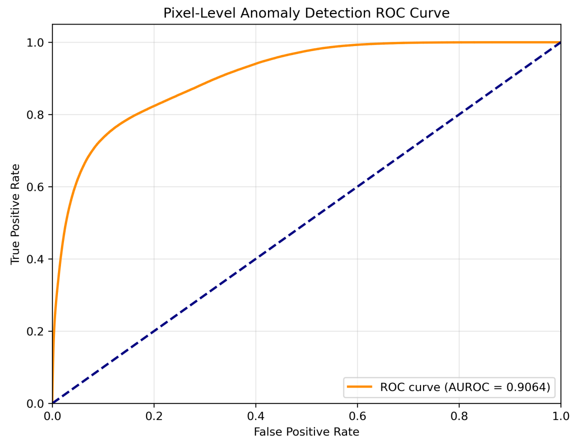

Fig. 1 Pixel-level ROC curve for anomaly localization (AUROC = 0.906). The model demonstrates strong ability to discriminate tumor pixels from healthy tissue based on reconstruction error

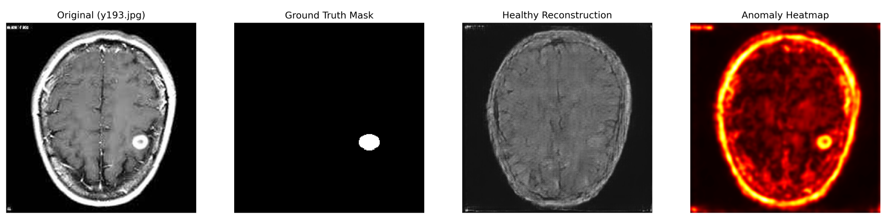

Fig. 2 Qualitative anomaly detection example. Left to right: original brain MRI with tumor, ground truth tumor mask, healthy reconstruction, and anomaly heatmap. High-intensity regions in the heatmap correspond to areas where the healthy reconstruction deviates from the original, successfully localizing the tumor

5.2 Hyperparameter Ablation Study

We searched 120 configurations: 10 checkpoints (50K–140K steps) × 4 noise levels ($t_{\text{start}} \in \{150, 200, 250, 300\}$) × 3 guidance scales ($w \in \{3.0, 5.0, 7.5\}$). Table 3 presents the top 5 configurations.

Effect of Noise Injection Level (t_{start}).

This emerges as the most critical hyperparameter. Table 4 shows that AUROC peaks at 0.756 for $t_{\text{start}} = 150$ and drops to 0.649 at $t_{\text{start}} = 400$, confirming that excessive noising destroys healthy features while insufficient noising fails to remove the anomaly signal.

Table 3 Top 5 hyperparameter configurations by image-level AUROC (grid search over 120 combinations)

Checkpoint	t_{start}	w	**AUROC**	**F1**	**Sens.**	**Spec.**
step_100K	150	7.5	**0.756**	0.784	0.784	0.564
step_80K	200	7.5	0.755	0.754	0.711	0.649
step_100K	150	3.0	0.755	**0.820**	**0.880**	0.467
step_140K	150	3.0	0.755	0.785	0.784	0.573
step_100K	150	5.0	0.755	0.742	0.689	0.662

Table 4 Effect of noise injection level t_{start} on anomaly detection (step_100K, w = 7.5)

t_{start}	**AUROC**	**F1**	**Sensitivity**	**Specificity**
150	**0.756**	0.784	0.784	0.564
200	0.749	0.767	0.738	0.627
250	0.739	0.793	0.811	0.529
400	0.649	0.539	0.402	0.818

Effect of Guidance Scale (w).

AUROC remains stable across guidance values (varying <0.002), but w modulates the sensitivity–specificity trade-off. At $w = 3.0$, sensitivity reaches 0.880 with specificity 0.467; at $w = 7.5$, the balance shifts to 0.784/0.564. Stronger guidance produces more conservative reconstructions, reducing false positives.

Effect of Checkpoint Maturity.

Checkpoint 100K consistently produces the best AUROC. Earlier checkpoints (50K–90K) show slightly lower AUROC (incomplete convergence); later checkpoints (110K–140K) show comparable but marginally reduced performance, suggesting mild overfitting.

5.3 Synthetic Image Quality

Table 5 evaluates 500 synthetic images generated with DDIM sampling (50 steps, η = 0, healthy class).

The FID of 144.43 reflects moderate quality, contextualized by our small training set (~1,500 images) and limited calibration of ImageNet-based FID for grayscale medical imagery. The diversity score (LPIPS = 0.404) confirms varied generation without mode collapse.

Table 5 Synthetic image generation quality metrics (500 generated images vs. real healthy test images)

Metric	Value
FID ↓	144.43
SSIM ↑	0.129 ± 0.085
LPIPS ↓	0.534 ± 0.033
Diversity (LPIPS) ↑	0.404 ± 0.104

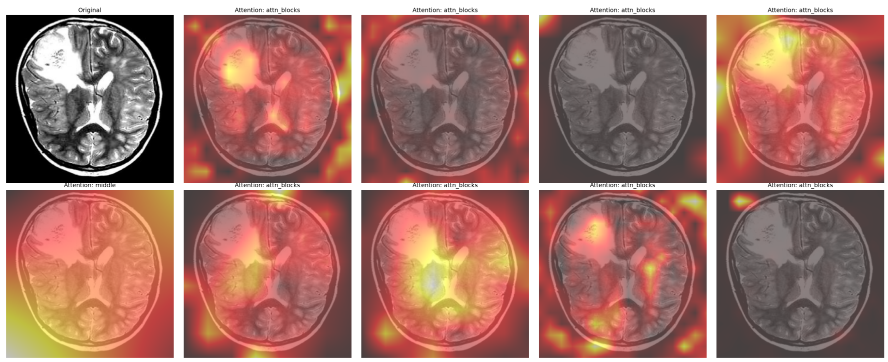

Fig. 3 Self-attention maps from the U-Net during denoising. The model attends to anatomically relevant structures, demonstrating learned spatial awareness of brain morphology

5.4 Explainability Analysis

Figure 3 visualizes self-attention weights from the U-Net at resolutions 16×16 and 8×8, revealing that the model focuses on anatomically meaningful structures—ventricular boundaries, cortical folds, and tissue interfaces. Figure 4 compares denoising trajectories: for healthy inputs, reconstruction converges smoothly; for anomalous inputs, the model progressively "corrects" the tumor region, replacing it with healthy tissue.

6 Discussion

Anomaly Detection Mechanism.

Our results confirm that latent diffusion models can detect brain tumors through reconstruction-based scoring without anomaly labels. The DDPM, trained only on healthy anatomy, "corrects" pathological regions during healthy-conditioned reconstruction, producing measurable error in tumor areas.

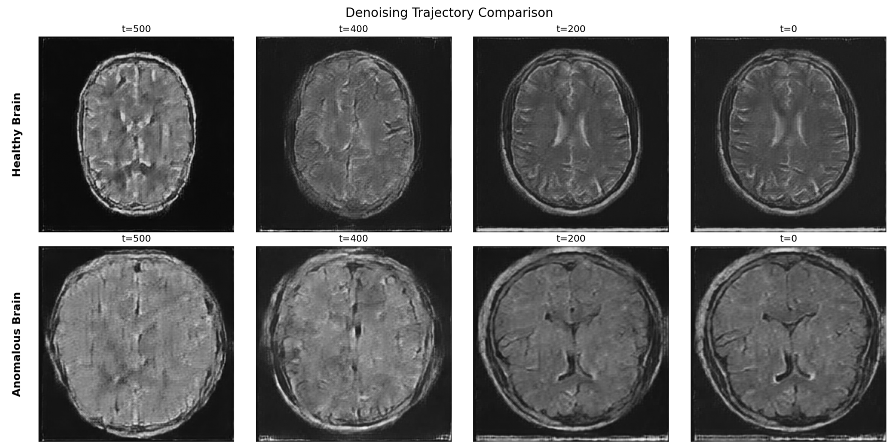

Fig. 4 Denoising trajectory comparison between a healthy brain MRI (top) and a tumor-bearing MRI (bottom). The healthy-conditioned reconstruction progressively "corrects" the tumor region

Critical Role of t_{start}.

The ablation study identifies t_{start} as the dominant factor in detection performance. Sufficient noise must destroy the anomaly signal, but excessive noise also destroys healthy features, causing hallucinated anatomy and false positives. Our optimal $t_{\text{start}} = 150$ ($T = 1000$) balances these objectives, consistent with findings in AnoDDPM [25].

Guidance Scale as a Clinical Tuning Knob.

The guidance scale w provides a sensitivity–specificity trade-off with minimal AUROC impact. At $w = 3.0$, the model achieves 88% sensitivity at the cost of more false positives; at $w = 7.5$, the balance shifts toward specificity. This tunability is valuable for clinical deployment where cost asymmetries between missed tumors and false alarms vary by application.

Generation Quality and Comparison.

The FID of 144.43 should be interpreted considering the small training set (~1,500 grayscale images) and ImageNet-calibrated metrics. Our image-level AUROC of 0.756 falls within ranges reported by comparable methods: AnoDDPM [25] achieves 0.70–0.83 and Wolleb et al. [24] report 0.74–0.86. Our pixel-level AUROC of 0.894 is competitive despite operating at coarse latent resolution (32×32) upsampled to 256×256.

Inference Efficiency.

With CFG and $S = 50$ DDIM steps, each pass requires $2S = 100$ U-Net evaluations. Operating in the 32×32 latent space reduces FLOPs by ~50× versus pixel space, and DDIM's 20× step reduction yields ~1000× overall speedup. Inference completes in ~300–350 ms per image on an RTX 4060, with ~52M total parameters (35M U-Net + 17M VAE).

Limitations.

(1) **2D processing** ignores volumetric context. (2) The **limited dataset** (~3,000 images from a single source) constrains anatomical diversity. (3) **No demographic metadata** precludes fairness analysis. (4) The 8× VAE compression creates a **resolution bottleneck**, limiting localization to ~8×8 pixel patches. (5) **No clinical validation**—deployment requires prospective validation with radiologist assessment.

Ethical and Regulatory Considerations.

Diffusion models trained on small datasets risk memorizing individual examples [1, 5]; mitigations include perceptual similarity monitoring and differential privacy. The absence of demographic metadata precludes fairness analysis. Synthetic images must carry provenance labels and not substitute for real diagnostic data. Clinical application requires compliance with the EU AI Act and FDA SaMD guidance. This work is a research prototype and is **not suitable for clinical diagnosis**.

7 Conclusion

We have presented a two-stage latent diffusion pipeline combining a MONAI-based VAE with a class-conditional U-Net DDPM for brain MRI anomaly detection and synthetic image generation. The system achieves an image-level AUROC of 0.756 and pixel-level AUROC of 0.894, competitive with existing diffusion-based methods. Our ablation across 120 configurations reveals that t_{start} dominates detection performance, while the guidance scale provides a clinically useful sensitivity–specificity trade-off. The model generates diverse synthetic brain MRIs (FID = 144.43). We further contribute explainability mechanisms and discuss ethical considerations. Future work includes 3D volumetric processing, larger multi-institutional datasets, multi-modal conditioning, and prospective clinical validation.

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RTX 4060 GPU, demonstrating the accessibility of latent diffusion methods for medical imaging research.

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